

CADENCE AI™

End-to-End Platform Architecture

How Cadence AI crawls, thinks, learns, generates, and iterates. A deep technical breakdown of the onboarding engine, the neural reasoning pipeline, the strategic pattern library, and the execution engine.

5

INTELLIGENCE LAYERS

2

AI REVIEW STAGES

21MARKETING
PLAYBOOKS**100%**

AUTO ONBOARDING

PHASE 1: DEEP INTAKE

The Zero-Friction Onboarding Engine

Users don't manually answer 50 questions about their brand. They just provide a URL. Cadence AI uses headless browsing and multi-modal processing to construct a semantic "Brand OS."



Jina AI Web Crawler

Bypasses anti-bot protections to scrape down the target site. Converts messy HTML into clean, semantically rich Markdown context.



Vision & OCR Pipeline

Processes product images and screenshots using OCR and Vision models to extract deep context beyond just the written website copy.



Tavily Deep Research

Runs autonomous background searches to find competitors, external reviews, and market sentiment to cross-reference brand claims.

Outputs: The Brand OS

- ◆ **Tone Fingerprint:** Auto-detects the brand's exact tone (e.g. "Irreverent, punchy, anti-corporate").
- ◆ **Target Personas:** Identifies psychometrics, not just demographics.
- ◆ **Core Pillars:** Isolates the 3-5 unique value props to hit in content.

```
// Example Intake Execution Log  
Starting Deep Crawl: https://founder-site.com  
Jina Reader: Parsed 18,204 words of Markdown...  
Vision AI: Extracted context from 4 hero graphics...  
Tavily: Found 3 direct competitors...  
Result: Generated Living Brand OS successfully.
```

PHASE 2: CORE KNOWLEDGE

The Static Playbook Brain

Unlike generic AI tools that guess at marketing, Cadence is trained on a proprietary, server-side knowledge base of 21 deep-dive digital marketing playbooks — covering every platform, algorithm, and growth tactic.



Platform Algorithms

Instagram, TikTok, YouTube, LinkedIn, X — each with dedicated playbooks covering ranking signals, formats, and retention mechanics.



Creative & Copywriting

Persuasion psychology, viral hook frameworks, tone calibration, and strict anti-"AI smog" guardrails protecting every word.



Analytics & Experimentation

A/B testing protocols, statistical significance thresholds, and incrementality models built into the foundation.

PHASE 3: THE FRAMEWORKS

The Strategic Pattern Vault

47 battle-tested positioning and narrative frameworks. The AI doesn't just "write content" — it intentionally executes a strategic warfare playbook, enforced aggressively.

Positioning Patterns (How to Win)

- ◆ The Category Creator — Own a new market
- ◆ The Anti-Thesis — Attack the status quo
- ◆ The Controversial Truth — Lead with polarizing facts
- ◆ The Enterprise Trojan Horse — Bottom-up wedges

Narrative Patterns (How to Speak)

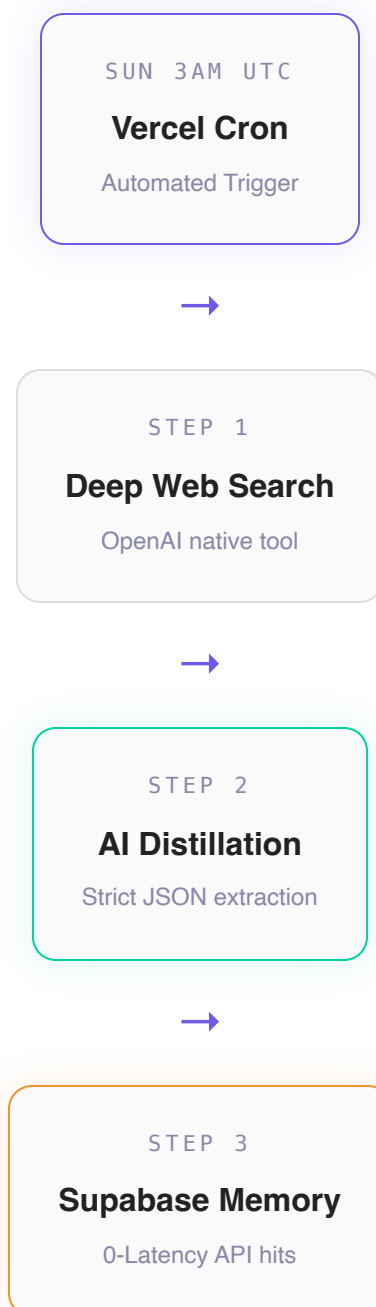
- ◆ The Founder's Raw Journal — Zero polish
- ◆ The Data Assassin — Only heavy statistics
- ◆ The Myth Destroyer — Dismantle industry lies
- ◆ ... and 40 more patterns encoded.

```
// Each pattern enforces mandatory guardrails:  
Pattern selected: "The Anti-Thesis"  
Markers: ["Name enemy explicitly", "Comparison tables"]  
Banned Words: ["innovative", "cutting-edge", "best-in-class"]  
AI Engine is structurally prevented from ignoring these limits.
```

PHASE 4: LIVE DATA

Live Algorithm Scout

A fully automated serverless pipeline that scrapes the live web weekly to discover and verify organic algorithm changes — injecting them directly into the context stack.





Anti-Guru Filter

Only ingests rules implicitly backed by verified data sets or official platform announcements. No speculation.



Failure Resilience

If web search or distillation fails, existing cache is NEVER overwritten. We never break the AI due to upstreams.

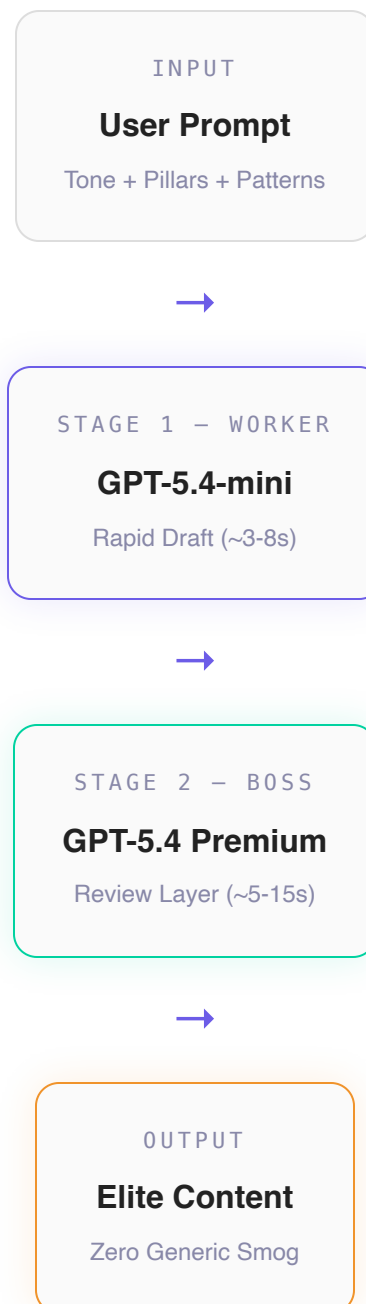


Zero Latency

Live rules are pulled from Supabase via `pgvector` & global cache with sub-30ms read paths for instantaneous logic.

The Two-Tier Reasoning Pipeline

Every piece of complex content undergoes a dual-pass evaluation. A fast "Worker" drafts it comprehensively. A ruthless "Boss" reviews, edits, and kills "AI smog."





The Worker (Stage 1)

Hits the ground running. Aggregates the huge context stack (Playbooks, Brand OS, Tone, Audience) and builds a structurally sound draft rapidly.



The Boss (Stage 2)

The "Ruthless Creative Director". Responsible entirely for tone lock, cultural nuances, and completely banning ChatGPT-isms like "Unlock your potential," "Elevate," or "Seamless."

PHASE 6: EXECUTION

Strategic Calendar Mapping

The final output isn't a long block of text. The "Premium Single-Tier" orchestration model transforms the raw intelligence into a robust, day-by-day JSON calendar interface.



1. One-Shot Full Month Mapping

Instead of generating post-by-post, the Premium layer receives the full Brand OS, the Live Algorithms, and the 4-week context to guarantee narrative sequencing. It ensures a story evolves naturally from Week 1 to Week 4.



2. Cross-Platform Nuance

The AI isolates content by platform context. A Tuesday post is split natively: the LinkedIn draft leans heavily on textual data, while the Instagram output structures itself for a visual Carousel hook.



3. Content Ratio Automation

Autonomously balances *Top of Funnel* (high-reach hooks), *Middle of Funnel* (deep-dive value), and *Bottom of Funnel* (direct response conversion) across the UI grid.



4. Immediate UI Injection

Strict schema outputs (Response Format JSON) flow directly into our Next.js + React drag-and-drop dashboard for immediate human approval, avoiding parsing errors seamlessly.

PHASE 7: REFINEMENT

The "Ask AI" Creative Director Chat

Generation is never perfect on the first try. Instead of forcing users to write complex prompts to fix issues, we built an iterative chat interface anchored specifically to existing content blocks.

- Context-Aware Refinement:** The AI already holds the Brand OS, Tone Fingerprint, and Calendar strategy. When a user says "Make this angrier," it applies the attitude shift *without breaking the brand positioning*
- Live Diff Injection:** Updates are passed directly back into the UI state machine, so users see edits instantly on their calendar canvas.
- Actionable Suggestions:** If users are stuck, the Boss AI proactively offers alternative strategic angles (e.g., "Pivot this to a Myth Destroyer pattern").
- Vercel AI SDK Integration:** Fully streamed conversational responses for zero-latency feeling.

```
// Standard Prompts failing:
User: "Make it funny" (Gets generic dad jokes)

// The Creative Director Mode:
System: "Refining Draft ID_3914. Applying Brand OS Tone: 'Cynical but helpful'.
Applying Pattern: 'The Data Assassin'."
User: "Punch up the hook."
Boss AI: "Rewritten. Added statistics. Eliminated generic intro."
```

DEEP DIVE

The Neural Context Stack

Every time Cadence hits the LLM, this is the exact knowledge hierarchy injected into the Context Window — ordered from highest to lowest priority.

PRIORITY	CONTEXT LAYER	SIZE	SOURCE	UPDATE FREQUENCY
P0	Live Algorithm Intelligence	~500 tokens	Supabase (cached)	Weekly (automated)
P1	Strategic Pattern Vault	~25,000 tokens	Markdown (server)	Manual (curated)
P2	21 Marketing Playbooks	~20,000 tokens	Markdown (server)	Manual (curated)
P3	Brand-Specific OS	~2,000 tokens	User-generated	Per brand (dynamic)
P4	User Prompt & Task	~200-1000 tkns	User session	Per request

```
// With OpenAI Prompt Caching: 90% discount on static layers (P1 + P2)
Total Context In: ≈ ~47,700 tokens
Total Generation Out: ~300 – 2,500 tokens
```

Token Economics & Overhead

Built for scale. Cost mapping for the orchestration pipeline utilizing GPT-5.4 pricing tiers alongside OpenAI's caching.

FEATURE COMPONENT	MODEL DEPLOYED	AVG INPUT TOKENS	AVG OUTPUT TOKENS	EST. COST / RUN
Auto Brand Research	5.4-mini	~25,000	~1,000	\$0.02
Strategy Core Generation	5.4 Premium Boss	~47,000	~2,000	\$0.20 – \$0.35
Full Monthly Calendar	5.4 Premium	~23,000	~2,500	\$0.09
Single Content Draft	2-Stage Engine	~42,300	~600	\$0.07
Director Chat Edit	5.4-mini	~21,000	~200	\$0.01

\$2.10

FULL MONTH OF CONTENT (X30)

90%

SAVINGS VIA PROMPT CACHE

\$1000+

AGENCY EQUIVALENT VALUE

SUMMARY

The Fully Autonomous SaaS Stack

Cadence AI is not an OpenAI wrapper. It is a purpose-built, structurally robust reasoning engine simulating a \$15,000/mo elite creative agency.

1. Automated Knowledge

Crawls brand URLs natively using Jina and Vision models to extract deep intelligence with zero user effort.

2. Static & Live Rules

Forces the AI to obey 21 marketing playbooks, 47 positioning patterns, and live scraped algorithm findings natively.

3. The Boss Editor

Prevents AI language hallucination natively. A secondary orchestrator that actively hunts and deletes generic terminology.

4. Strategic JSON Mappings

Maps highly abstract marketing thoughts into 30-day concrete execution calendars via single-shot Premium reasoning models.

5. Creative Chat Loops

Empowers users with a Vercel-SDK conversational interface to tweak drafts cooperatively with their dedicated digital creative director.

